Imad Ahmad - MLOps Engineer

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SUMMARY

Machine Learning Engineer specializing in MLOps and production ML systems on GCP and Vertex AI, with hands-on expertise architecting Kubeflow pipelines, CI/CD-driven model deployment, and automated monitoring across the full ML lifecycle. Proven track record of building end-to-end deployments for mission-critical data infrastructure, scaling production models to handle millions of daily records with 99.9% availability. Published statistical researcher who combines software engineering best practices with advanced theoretical knowledge to build robust, high-performance systems for complex environments.

EXPERIENCE

TELUS Communications Inc. **October 2024 – Present**

Machine Learning Engineer - MLOps, *July 2025 - Present*

- Productionized a PyTorch deep learning churn model end-to-end on **GCP Vertex AI** using **Kubeflow pipelines** spanning training, validation, versioning, deployment, and monitoring across millions of customer records - surfacing insights leveraged by **director-level leadership** to shape proactive retention strategy.
- Engineered scalable, **automated ML pipelines** on GCP Vertex AI processing 1M+ daily records by integrating Kubeflow orchestration, CI/CD-driven deployment, automated error alerting, and production monitoring - delivering reliable daily predictions to stakeholders with minimal manual intervention.
- **Partnered with data scientists** to standardize the experimentation-to-production path on Vertex AI by defining reproducible deployment patterns, model versioning, and validation gates, while contributing modeling depth in gradient flow, optimization, and loss-function design to harden the accuracy and robustness of production systems.

Data Engineer, *October 2024 – July 2025*

- Spearheaded a cross-functional migration initiative involving multiple engineering teams to execute a **zero-downtime** transition of **100+ mission-critical ETL pipelines** to Google Cloud Composer, meeting SLAs for **99.9% data availability**.
- Architected and led the QA automation strategy for the **Telus-WestJet strategic partnership**, designing **Python-based data validation pipelines** that reconciled cross-company architectures and guaranteed **99.99% accuracy** in high-volume customer identity synchronization.
- Designed a **modular DAG generation framework** within Cloud Composer using **Python** and custom operators, allowing teams to deploy configuration-based pipelines without writing repetitive boilerplate code.

LifeLabs Inc. **August 2021 – July 2022**

Laboratory Data Assistant, *August 2021 – July 2022*

- Designed and deployed 15+ complex **Apache Airflow DAGs** to automate **critical ETL pipelines**, integrating **CI/CD workflows** to ensure version-controlled releases and system reliability - reducing overall data processing time by 40%.
- Optimized daily extraction and transformation of **1,000+ COVID-19 test records** through **high-performance SQL queries** and implementing observability via **Prometheus** and **Grafana**, improving operational reliability and decision-making.

EDUCATION

Master of Data Science and Analytics, *University of Calgary* **Graduated December 2023**

Bachelor of Science in Integrated Science, *University of British Columbia* **Graduated May 2020**

PROJECTS AND PUBLICATIONS

Doc-Shield – Production Ready SaaS Solution

https://doc-shield.com

- Designed and built "Doc-Shield," a secure web-based document redaction platform leveraging **Google Cloud** (Cloud Functions, Cloud Run, Firestore, GCS) to automatically redact personal information from uploaded documents.
- Led full lifecycle development from concept to deployment, including **backend orchestration**, **CI/CD integration**, **UX optimization**, and branding strategy, resulting in a **production-ready SaaS solution**.

Master's Degree Final Project – Recycling Image Classifying Robot

https://youtu.be/s5CwtBsv_bo

- Developed a **Convolutional Neural Network** in **TensorFlow** that classifies waste images with **85% accuracy**.
- Managed and analyzed over 10,000 labeled waste images with **Python** and the **OpenCV** library, sourced from **Kaggle** datasets and web scraping with **Selenium**, to utilize for training.
- Integrated the model into a Raspberry Pi robot, demonstrating the model's practical application in dynamic environments.

Academic Statistical Research (Published) – Exploring the Gender Disparity in Physiology Departments

https://pubmed.ncbi.nlm.nih.gov/33409087/

- Utilized advanced statistical methods in **Python** and **R** to explore the gender disparity in Physiology departments for over 2000 faculty members, contributing valuable insights into diversity dynamics within academic settings.
- Authored and **successfully published** research in a peer-reviewed journal as the **first author**, demonstrating proficiency in scientific communication and scholarly writing.

CERTIFICATIONS

GCP Cloud Digital Leader • 2024

https://www.credly.com/badges/72caaec7-b019-4bc1-a2e0-c7f2ad21b16a/public_url

AWS Certified Cloud Practitioner • 2023

https://www.credly.com/badges/22444eaa-2cfa-4a7d-ade2-4608ca8922a0/public_url